

# Vaibhav Panchal

Junior AI/ML Developer | Deep Learning Researcher | Full-Stack AI Engineer  
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## PROFESSIONAL SUMMARY

AI/ML Developer at TestGrid Labs Inc. building intelligent test automation pipelines using deep learning and LLMs. M.E. in Software Engineering (LDCE, GTU, CGPA 8.21/10). Research work in Computer Vision (YOLOv11 + ECA, +3.8% mAP50 on DIOR-R, under review) and Cybersecurity (hybrid IDS, 98.4% accuracy, unpublished manuscript). Hands-on across the full ML lifecycle from data engineering and model training to Flask deployment and CI/CD integration. Core stack: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LangChain, Flask, Nginx, AWS, Docker.

## TECHNICAL SKILLS

**Languages:** Python (Advanced), SQL, JavaScript, C/C++, Java, HTML/CSS  
**Frameworks and Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, XGBoost, LangChain, LangGraph, FastAPI, Flask, Streamlit  
**AI and ML:** Computer Vision, NLP, LLMs, RAG, Object Detection, Prompt Engineering, OpenAI API, Ollama, RAGAS, Langfuse  
**Cloud and Infrastructure:** AWS (EC2), Docker, Nginx  
**Databases:** Redis, SQLite, ChromaDB, Pinecone (Vector DB)  
**Tools and Practices:** Git, Jupyter, MLflow, Postman, Selenium Grid, Bitbucket, REST API Design, CI/CD

## WORK EXPERIENCE

**Junior AI/ML Developer** | TestGrid Labs Inc. | Navsari, Gujarat May 2026 - Present

- Built deep learning inference pipelines for automated UI regression detection, reducing manual test triage time by flagging anomalies before human review.
- Designed and deployed Flask-based model-serving endpoints integrated into the CI/CD pipeline, enabling real-time ML inference on test runs at scale.
- Prototyped an LLM-assisted test case generation module using the OpenAI API, reducing repetitive test authoring effort across the QA team.
- Configured Nginx as a reverse proxy in front of Flask model-serving endpoints, handling request routing and load distribution for production ML inference traffic.

**Trainee Associate Software Developer** | Techreale Pvt. Ltd. | Valsad, Gujarat Jan 2024 - Apr 2024

- Delivered production features for the Famyll healthcare platform (React.js, Node.js, MongoDB) across claim management, auth flows, and real-time communication modules.
- Reduced API response handling bugs by refactoring async integration patterns; contributed across 3 sprint cycles in an Agile team of 8.

**Summer Intern, Frontend Development (React)** | INFOLABZ IT Services Pvt. Ltd. | Ahmedabad, Gujarat Jul 2023

- Built and optimized React.js dashboard components with async API integration; improved rendering performance through targeted component-level refactoring.

## PROJECTS

**Multi-Agent Customer Support System - LangGraph Orchestration** 2026  
*Python | LangGraph | LangChain | FastAPI | Redis | Langfuse*

- Built a multi-agent customer support platform with a Supervisor Agent that classifies query intent and routes to specialized Billing, Technical Support, and Refund sub-agents implemented as LangGraph state-graph nodes.
- Preserved context across agent hand-offs via summarized state transfer and persisted conversation memory (Redis + SQLite), so a customer's session survives across requests without repeating themselves.
- Instrumented every agent call, tool call, and hand-off with Langfuse tracing for full observability into how a single ticket moved through the system.

**Production RAG System with Evaluation** 2026  
*Python | LangChain | ChromaDB/Pinecone | RAGAS | Streamlit*

- Built a RAG pipeline with hybrid retrieval (dense vector similarity + BM25 keyword search) and a citation-enforced grounding prompt to reduce hallucination risk.
- Built a first-class evaluation suite using RAGAS metrics (Faithfulness, Context Precision, Answer Relevancy), benchmarking every prompt or retrieval change against a held-out Q&A test set before it's considered shipped.
- Tracked evaluation results over time with Langfuse to catch quality regressions before they reached users, surfaced via a live Streamlit metrics dashboard alongside the chat interface.

ExamForge - LLM-Powered Exam Question Generator

2026

Python | LangChain | Ollama | OpenAI GPT | Streamlit

- Built an end-to-end RAG pipeline: ingests syllabus PDFs, chunks and embeds content, then prompts an LLM to generate structured question papers with configurable difficulty and Bloom's taxonomy tagging.
- Supported both local inference (Ollama) and cloud (OpenAI API), making it deployable in air-gapped institutional environments; cut manual question-drafting time by ~80% in pilot use.
- Engineered structured JSON output parsing with validation to ensure generated questions conform to a strict schema before rendering to PDF.

Intrusion Detection System - CNN + BiLSTM + MultiHead Attention

Apr - May 2025

Python | TensorFlow/Keras | Scikit-learn | PCA | AutoEncoder

- Designed a hybrid CNN-BiLSTM-Attention architecture for multi-class network traffic classification; reached 98.4%+ accuracy on NSL-KDD and CICIDS benchmarks, outperforming Random Forest and SVM baselines.
- Applied PCA and a stacked AutoEncoder for feature compression, reducing input dimensionality by ~60% without accuracy loss, improving inference speed for real-time deployment scenarios.
- Structured the codebase as a modular pipeline (ingest, preprocess, train, evaluate) to support easy retraining on new traffic datasets.

Content-Based Movie Recommendation Engine

Jan - Feb 2025

Python | Scikit-learn | TF-IDF | Cosine Similarity | Streamlit

- Implemented TF-IDF vectorization and cosine similarity on the TMDb 5000 dataset; deployed as a real-time Streamlit app with sub-second recommendation latency.
- Pipeline is architected for extensibility - collaborative filtering and hybrid approaches can be plugged in without restructuring the serving layer.

RESEARCH WORK

Channel Attention for Oriented Object Detection in Remote Sensing

2025 (Under Review)

YOLOv11-OBb | ECA Attention | DIOR-R Dataset | PyTorch

- Integrated Efficient Channel Attention (ECA) into YOLOv11s-OBb to improve feature recalibration for small and densely packed objects in aerial imagery.
- Evaluated on DIOR-R (20-class remote sensing benchmark): achieved +3.8% mAP50 and +4.3% recall over the baseline, with no increase in parameter count.

Hybrid Deep Learning Architecture for Network Intrusion Detection

2025 (Manuscript in Preparation)

CNN + BiLSTM + Attention | NSL-KDD | CICIDS | TensorFlow

- Proposed a sequential CNN-BiLSTM-Attention IDS that captures both spatial and temporal patterns in network flow data; applied PCA and AutoEncoder for feature compression.
- Achieved 98.4%+ accuracy on NSL-KDD and CICIDS; outperformed Random Forest and SVM across all attack categories on precision, recall, and F1-score.

EDUCATION

M.E. - Software Engineering - GTU, LDCE, Ahmedabad

2024-2026 | CGPA: 8.21/10

B.E. - Computer Engineering - GTU, GDEC, Navsari

2020-2024 | CGPA: 8.11/10

ACHIEVEMENTS & CERTIFICATIONS

- SSIP Hackathon 2022 - Developed an automated smart attendance management system; selected among top teams from Gujarat.
- Dev-o-lution Hackathon - Certificate of Participation for building a competitive solution under a 24-hour constraint.
- 2 research works in Computer Vision (remote sensing, +3.8% mAP50, under review) and Cybersecurity (IDS, 98.4% accuracy, manuscript in preparation).

LEADERSHIP & EXTRACURRICULAR

- Volunteered at college-level tech events and coding quiz competitions, supporting event coordination and participant engagement.

INTERESTS

AI/ML Research, LLMs & RAG Systems, Computer Vision, Deep Learning, MLOps, Automation, Cloud Computing, Data Science | Chess, Exploring Emerging Technologies, Music, Video Games